

Competitive Programming

Section 26: Tree DP Basics

Competitive Programming Course

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1 Why tree DP comes after interval DP

A tree has no cycles. Once you choose a root, every edge points from parent to child, and every subtree becomes an independent region after the parent's state is fixed.

Tree DP is the habit of asking:

If I know what happens at node u , what choices remain independently inside each child subtree

This section stays with one-root, bottom-up DP. Rerooting and LCA appear later.

2 Rooting a tree

The input tree is undirected. To do DP, pick any root, usually vertex 1.

After rooting:

```
parent[u] = previous vertex on the path to the root
children[u] = neighbors except parent[u]
subtree[u] = u plus all descendants of u
```

For large trees, avoid recursive DFS unless you deliberately handle stack limits. The local solutions build an iterative order:

```
order = [root]
for u in order:
    for v in adj[u]:
        if v has no parent:
            parent[v] = u
            append v to order

process reversed(order)
```

`reversed(order)` is postorder: every child is processed before its parent.

3 Recognition

Use tree DP when:

Statement signal	Likely state
choose vertices with parent-child conflict	include/exclude node
count colorings with adjacent restrictions	state by color of node
choose disjoint edges	matched/unmatched state
combine independent child subtrees	product or sum over children
answer depends on every root choice	future rerooting, not this section

4 Pattern 1: include/exclude node

For maximum independent set on a weighted tree:

```
take[u] = best total in subtree[u] if u is chosen
skip[u] = best total in subtree[u] if u is not chosen
```

If u is chosen, every child must be skipped:

```
take[u] = weight[u] + sum(skip[v])
```

If u is skipped, every child may be chosen or skipped:

```
skip[u] = sum(max(take[v], skip[v]))
```

Answer:

```
max(take[root], skip[root])
```

5 Pattern 2: counting colorings

For black/white colorings with no black-black edge:

```
white[u] = number of valid colorings of subtree[u] if u is white  
black[u] = number of valid colorings of subtree[u] if u is black
```

If u is white, each child can be either color:

```
white[u] = product(white[v] + black[v])
```

If u is black, every child must be white:

```
black[u] = product(white[v])
```

Use modulo after each multiplication.

6 Pattern 3: matching edges

For maximum matching in a tree, each vertex can be incident to at most one chosen edge.

A useful state:

```
free[u] = best matching in subtree[u] when u is not matched to any child
used[u] = best matching in subtree[u] when u may be matched to one child
```

First compute the base where u is not matched to a child:

```
base = sum(max(free[v], used[v])) over children v
free[u] = base
```

Then try matching u to one child v :

```
candidate = base
            - max(free[v], used[v])
            + free[v]
            + 1
```

Why `free[v]`? If edge (u,v) is chosen, vertex v cannot be matched to one of its children.

7 Worked example: include/exclude

Tree:

```
1(5)
  2(1)
  3(4)
    4(6)
    5(3)
```

Leaves:

```
take[2]=1, skip[2]=0
take[4]=6, skip[4]=0
take[5]=3, skip[5]=0
```

Node 3:

```
take[3] = 4 + 0 + 0 = 4
skip[3] = max(6,0) + max(3,0) = 9
```

Root 1:

```
take[1] = 5 + skip[2] + skip[3] = 14
skip[1] = max(1,0) + max(4,9) = 10
```

Answer: 14.

8 Worked example: colorings

For a path of three vertices:

1 - 2 - 3

Valid colorings are all $2^3 = 8$ colorings except those where adjacent black vertices appear:

BBB, BBW, WBB

So the answer is 5.

The DP gives the same result at the root.

9 C++ implementation notes

Use adjacency lists:

```
vector<vector<int>>> adj(n);
```

Build an iterative parent/order array. Process **order** backward. Use **long long** for weighted sums and modulo products.

10 Python implementation notes

Use `sys.stdin.buffer.read()` for large trees. Iterative traversal avoids recursion limits:

```
parent = [-1] * n
order = [0]
parent[0] = 0
for u in order:
    ...
for u in reversed(order):
    ...
```

11 Proof habits

Tree DP proofs use induction on subtree height.

Template:

```
Assume every child state is correct.  
Given u's state, child subtrees are independent.  
The transition combines exactly the allowed choices for children.  
Therefore u's state is correct.
```

Independence is the key tree property.

12 Common mistakes

```
Processing parent before children.  
Forgetting to exclude the parent from children.  
Using recursion and crashing on a chain.  
For include/exclude, allowing a chosen child under a chosen parent.  
For matching, matching u to more than one child.  
For coloring counts, adding child choices when they should multiply.
```

13 Local exercises

13.1 A. Festival Invite

Weighted independent set:

```
take[u], skip[u]
```

13.2 B. Road Pairing

Maximum tree matching:

```
free[u], used[u]
```

13.3 C. Tree Colorings

Count valid black/white colorings:

```
white[u], black[u]
```

14 Local judge workflow

Run:

```
CP_TARGET=solution python3 sections/26_tree_dp_basics/check.py
```

Run everything:

```
CP_TARGET=solution python3 tools/check_all.py
```

15 Submission checklist

```
Did I root the tree?  
Do I know parent and children for every node?  
Are children processed before parents?  
What exactly does each state mean?  
Do child contributions sum, multiply, or take max?  
Do I need modulo or long long?
```

16 Exercises

Root each tree before solving A, B, or C. State what a child contribution means, then test a leaf, a path, and a star to catch parent/child double counting.